

Introduction to Data Augmentation

- ✔ **Video:** Introduction to Data Augmentation
3 min
- ✔ **Practice Quiz:** Concept Check: Data Augmentation
Started
- ✔ **Reading:** Data Augmentation for Classification
15 min
- ✔ **Reading:** Data Augmentation Quick Reference
10 min

Data Augmentation for Object Detection

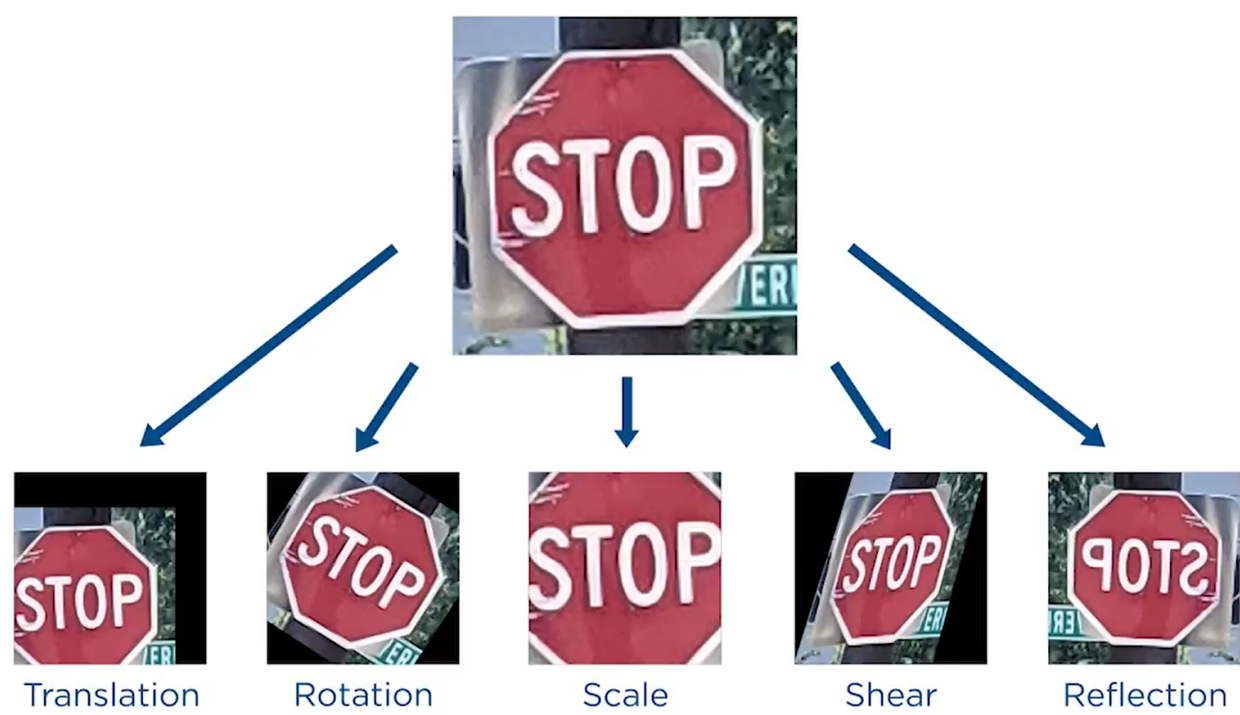
Data Augmentation Quick Reference

Data augmentation is a powerful technique if your model is not performing well and you suspect you lack enough variation in your images. It can save you significant time and effort collecting and labeling data. Data augmentation is also memory efficient because the synthetic images are used for training each epoch and immediately discarded. This saves lots of disk space, especially when training for 1000s of epochs.

However, it is usually best to collect more real images to avoid further biasing your models. For example, if all your images are taken on bright sunny days, using augmentation to simulate cloudy, foggy, or nighttime conditions is unlikely to capture these scenarios accurately.

Here are some tips when applying data augmentation:

1. There are five types of transformations covered in this course that you can use when augmenting your data: translation, rotation, scale, shear, and reflection.



2. Use your domain knowledge and application area to inform the types of transformations to use. For the stop signs above, reflection may not be appropriate. You may also set limits on how much rotation you add.

3. Increase your number of epochs to increase the variety of images. Since each epoch generates new augmented images, more epochs means more variety.

4. Only apply augmentation to the training images. The validation images are meant to serve as a check against overfitting. It is best to use real images for this purpose.

5. More types of augmentation are possible with custom functions. While beyond the scope of this course, other commonly used approaches are adding noise, adding motion blur, and adjusting contrast.

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